

Leicester's Healthcare and Education System: A Decade of Change (2012–2022)

A Policy-Oriented Data Analysis Using the UK Prosperity Index

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1. Introduction

Over the past decade, cities across the United Kingdom have experienced major social and economic changes shaped by austerity policies, demographic shifts, rising living costs, and the COVID-19 pandemic. Understanding how these long-term developments affect local public services is essential for policymakers, researchers, and local communities.

This project analyses trends in Leicester's healthcare and education systems between 2012 and 2022 using data from the Legatum Institute's *UK Prosperity Index 2022*. The aim of the analysis is to identify major changes in public well-being indicators and compare Leicester's performance with regional and national trends.

The study focuses specifically on healthcare and education because these two pillars showed some of the most significant long-term changes in Leicester during the analyzed period. While education indicators improved substantially, healthcare indicators demonstrated a noticeable deterioration over time.

The project combines descriptive statistical analysis with comparative regional analysis in order to better understand how Leicester has evolved relative to other major cities and the wider Midlands region.

2. Dataset and Methodology

2.1 UK Prosperity Index

The analysis is based on the *Going Deeper: UK Prosperity Index 2022* developed by the Legatum Institute. The index was designed to measure long-term wellbeing and prosperity across the United Kingdom.

The dataset includes:

- more than 225 indicators,
- 53 elements,
- 12 wellbeing pillars,
- over 10 years of historical data.

The pillars are grouped into three broad categories:

- Inclusive Societies,
- Open Economies,
- Empowered People.

Examples of analysed areas include governance, living conditions, safety, education, health, and investment.

2.2 Selection of Health and Education

An initial exploratory review of the dataset showed that several pillars, including Safety and Security, Personal Freedom, and Social Capital, experienced relatively small changes over time.

In contrast, Health and Education displayed substantial long-term variation. As a result, the analysis was narrowed to these two sectors.

The Health pillar consists of:

- Behavioural and Physiological Risk Factors,
- Preventative Interventions,
- Care Systems,
- Mental Health,
- Physical Health and Longevity.

The Education pillar consists of:

- Pre-primary Education,
- Primary Education,
- Secondary Education,
- Tertiary Education,
- Adult Skills.

Indicators are measured using percentages, rates, or population-adjusted values and combined into weighted scores.

2.3 Limitations

Several limitations should be acknowledged.

First, some historical observations in the Legatum dataset were repeated due to missing information. The years 2010–2012 showed noticeable duplication in several variables. For this reason, the analysis primarily focuses on the 2012–2022 period.

Second, the dataset identifies trends and changes in indicators but does not directly establish causation. Therefore, the analysis should be interpreted as descriptive rather than causal.

Finally, some sudden changes in indicators may partially reflect methodological updates, reporting differences, or disruptions related to the COVID-19 pandemic.

3. Healthcare Trends in Leicester

3.1 Overall Performance

Healthcare outcomes in Leicester declined considerably during the analysed period.

Compared with 2012 (baseline score: 52.6):

- Leicester's healthcare score decreased by approximately 16% - falling to 44.0 by 2022,
- healthcare performance fell below the national average,
- negative trends accelerated after the COVID-19 pandemic.

Although healthcare deterioration was visible across the entire country, Leicester experienced particularly visible pressures in preventative care and treatment efficiency.

3.2 Behavioural and Physiological Risk Factors

Several concerning health trends emerged within Leicester.

3.2.1 Childhood Obesity

Rates of obesity among young children increased significantly from 50% to 70% among 4–5-year-olds.

Rising childhood obesity may increase long-term risks of:

- diabetes,
- cardiovascular disease,
- reduced quality of life,
- future healthcare pressures.

At the same time, adult obesity indicators showed more volatile movements, suggesting possible changes in reporting methods or post-pandemic behavioural shifts.

3.2.2 Diet and Nutrition

Fruit and vegetable consumption declined from a stable 30% to between 10% and 20% in recent years.

Lower consumption of healthy foods may contribute to:

- obesity,
- poorer physical health,
- long-term chronic illnesses.

The trend may also reflect wider cost-of-living pressures affecting household food choices.

3.2.3 Smoking

Smoking prevalence increased from 30% before 2018 to 60% in 2022.

Smoking remains one of the leading preventable causes of:

- respiratory disease,
- cardiovascular illness,
- cancer,
- premature mortality.

The increase may indicate growing public health inequalities and worsening lifestyle-related health risks.

4. Preventative Healthcare

4.1 Cancer Screening

Participation in breast cancer screening programmes decreased from 60% in 2017 to 0% in 2022. Lower screening participation may reduce early detection rates and worsen long-term health outcomes. In contrast, bowel cancer screening participation improved from 10% before 2016 to 50% in 2022 - suggesting that some public health interventions achieved stronger engagement.

4.2 HPV Vaccination

HPV vaccination rates among girls declined from 70% in 2016 to just 40% in recent years. This trend is concerning because lower vaccination coverage may increase future risks of cervical cancer and other preventable diseases.

The decline may partly reflect:

- disruptions caused by the COVID-19 pandemic,
- vaccine hesitancy,
- reduced access to school-based health programmes.

5. Care System Efficiency

Healthcare system efficiency indicators deteriorated after the pandemic.

5.1 Emergency Care

The proportion of patients treated within four hours in emergency departments fell from a stable 50% before the pandemic to 40% in 2022.

This reflects broader NHS pressures including:

- staff shortages,
- increased demand,
- post-pandemic treatment backlogs.

5.2 Referral and Cancer Treatment Delays

Referral treatment performance dropped from 60% before COVID-19 to 20% in 2022. Cancer treatment performance declined gradually from 60% before 2014 to 20% by the end of the period. These findings suggest increasing pressure on healthcare capacity and resource allocation.

6. Regional Comparison: Healthcare

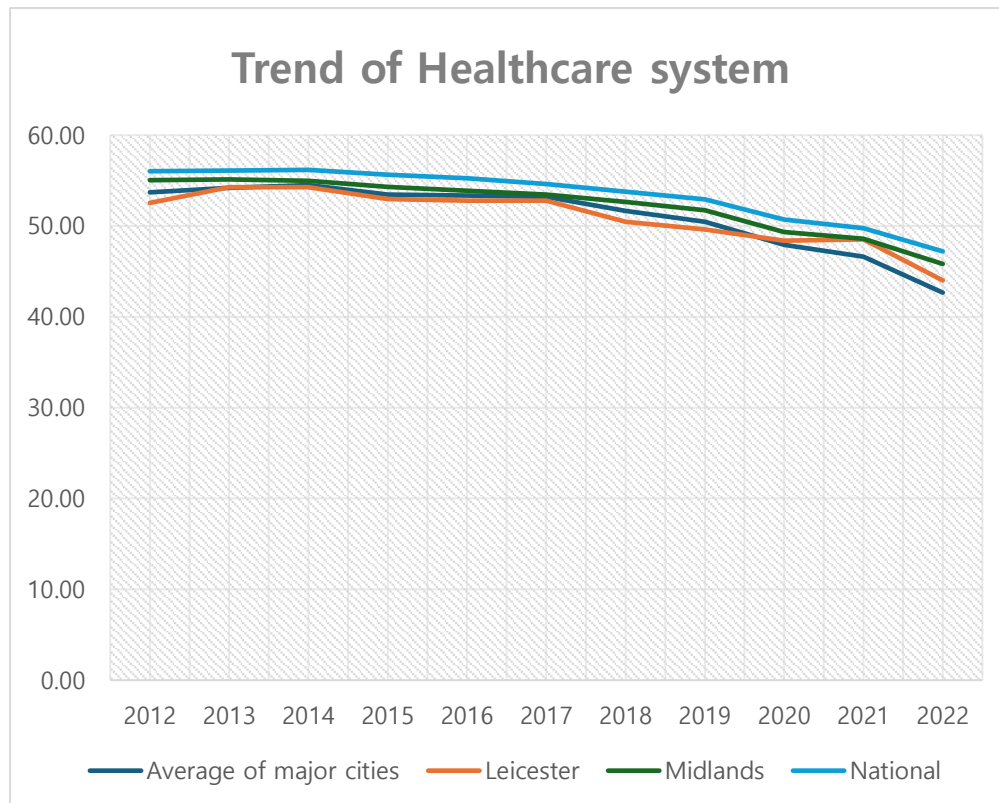
Leicester's healthcare performance was compared with:

- Nottingham,
- Derby,
- Coventry,
- Bristol,
- Stoke-on-Trent,
- the Midlands average,
- the national average.

During the early years of the dataset, Leicester performed relatively similarly to other Midlands cities. However, after 2014, healthcare indicators declined across nearly all regions. The COVID-19 pandemic accelerated this deterioration nationwide. Although Leicester did not experience the worst decline among Midlands cities, the city continued to perform below stronger-performing areas such as Bristol.

The comparison suggests that Leicester's healthcare challenges reflect both:

- local structural issues,
- wider national pressures affecting the NHS.



7. Education Trends in Leicester

7.1 Overall Performance

In contrast to healthcare, Leicester's education indicators improved significantly over the analysed period.

Compared with 2012:

- education scores increased by approximately 55–59%,
- Leicester eventually outperformed several comparable Midlands cities,
- adult skills indicators improved substantially.

The education system demonstrated strong long-term progress despite temporary stagnation between 2015 and 2020.

8. Primary and Secondary Education

8.1 Primary Education

Low-income primary attainment (literacy) increased from 38.7% in 2012 to 62.8% in 2022; numeracy increased from 38.6% to 70.6%.

This suggests improvements in:

- educational support,
- school performance,
- attainment among disadvantaged pupils.

8.2 Secondary Education

Secondary school outcomes also improved. Secondary attainment (Level 2 English and maths) rose from 30.3% in 2016 to 62.3% in 2022.

Improved attainment may positively influence:

- future employment opportunities,
- social mobility,
- long-term regional productivity.

9. Adult Skills and Lifelong Learning

One of the strongest improvements was observed in adult skills. Adults with any qualifications increased from 10% in 2012 to 50% in 2022; adults with at least Level 4 qualifications grew from 21.1% to 36.6%.

This may reflect:

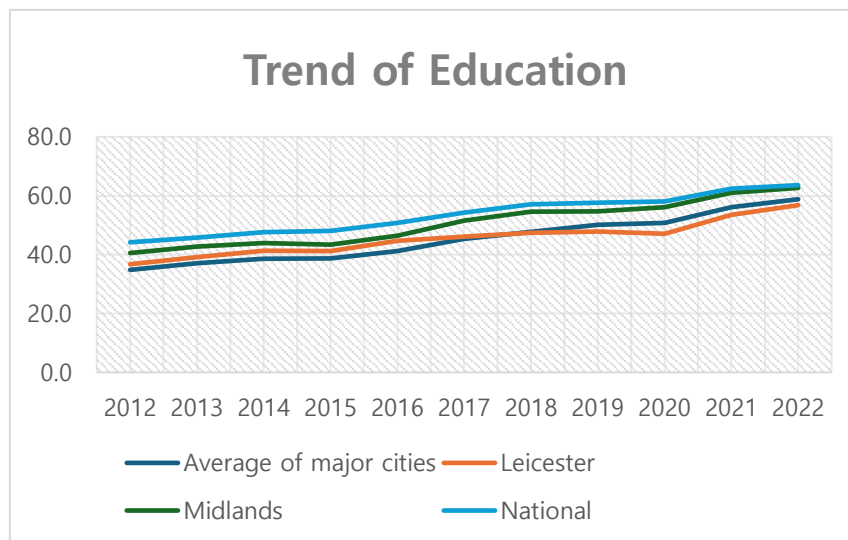
- expanded adult education opportunities,
- workforce retraining initiatives,
- growing demand for higher qualifications in the labour market.

The trend is particularly important because adult education contributes to:

- employability,
- economic resilience,
- regional development.

10. Regional Comparison: Education

Leicester's educational performance was compared with other major Midlands cities and national averages. Although Leicester initially performed below the national average, the city experienced strong long-term improvement. Between 2015 and 2020, educational progress slowed relative to faster-growing cities such as Derby and Stoke-on-Trent. However, after 2021, Leicester experienced a substantial acceleration in educational performance, consistent with broader national improvements. By the end of the period analyzed, Leicester demonstrated considerably stronger education outcomes than at the beginning of the decade.



11. Discussion

The analysis reveals a clear divergence between healthcare and education outcomes in Leicester. While education indicators improved significantly over the decade, healthcare outcomes deteriorated. Several broader explanations may contribute to this divergence.

Healthcare systems across the UK experienced:

- growing demand,
- post-pandemic backlogs,
- staffing pressures,
- rising chronic illness,
- increasing public health inequalities.

In contrast, educational improvements may reflect:

- long-term investment in schools,
- improved attainment strategies,
- expansion of adult skills programmes,
- increased emphasis on qualifications and employability.

The COVID-19 pandemic appears to have intensified existing healthcare pressures while having a less severe long-term effect on educational performance indicators.

12. Conclusion

This project examined long-term trends in Leicester's healthcare and education systems using data from the UK Prosperity Index.

The findings show that:

- healthcare outcomes deteriorated significantly between 2012 and 2022,
- preventative healthcare and treatment efficiency worsened,
- education indicators improved substantially,
- adult skills and educational attainment increased over time.

Although Leicester's healthcare challenges reflect wider national trends, the findings highlight the importance of continued investment in:

- preventative healthcare,
- public health interventions,
- healthcare capacity,
- educational development,
- adult learning opportunities.

The analysis demonstrates how regional data can be used to better understand long-term social change and identify areas requiring future policy attention.

References

- Legatum Institute – UK Prosperity Index 2022
- NHS and public health indicators
- Regional comparative statistics from the Midlands and UK national averages